

NITIN MEHRA

Backend Developer | Generative AI Enthusiast

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PROFESSIONAL SUMMARY

Computer Science undergraduate (final year B.Tech) with a solid foundation in Python, Java, SQL, and OOP, and hands-on experience building full-stack projects end to end. Built a project that lets you chat with a GitHub codebase and get AI-generated answers based on the actual code, along with a Python tool that automates cleaning and preparing raw datasets. Comfortable picking up new tools and frameworks quickly and enjoys building small working projects to actually learn something rather than just watching tutorials.

TECHNICAL SKILLS

Languages: Python, Java, C/C++

Generative AI / NLP: LLM APIs (OpenAI, Gemini), LangChain, Prompt Engineering, Hugging Face, RAG, Vector Embeddings

Backend & APIs: Spring Boot, Spring Security, REST APIs, JWT & OAuth2

Data & Databases: Pandas, NumPy, MySQL, MongoDB, Redis, Data Cleaning & Profiling

Tools & Platforms: Git, GitHub, Docker, Postman, AWS, Linux, VS Code

PROJECTS

RAG-Based GitHub Code Intelligence System [\(GitHub\)](#)

Stack: Java, Spring Boot, JPA, GitHub OAuth, Vector Search / RAG, Server-Sent Events (SSE)

- Built a full-stack platform that connects to a user's GitHub account via OAuth and syncs their repositories for AI-powered analysis, using Spring Boot and JPA on the backend.
- Designed a code-indexing pipeline that parses and chunks repository source files, converts them into vector embeddings, and stores them for semantic retrieval — so questions are answered using real code context instead of guesses.
- Implemented a Retrieval-Augmented Generation (RAG) chat backend with session management and Server-Sent Events (SSE), streaming AI answers to the frontend in real time instead of waiting for a full response.
- Built a chat interface that renders streaming markdown responses with inline citations, letting a developer ask natural-language questions about an unfamiliar codebase and see exactly which files the answer came from.

AutoClean-CSV — Automated CSV Data Cleaner & Profiler [\(GitHub\)](#)

Stack: Python, Pandas, NumPy, Object-Oriented Design

- Built a modular, production-style OOP pipeline (CSVCleaner + CSVProfiler) that automates the extract-transform stage of ETL — handling missing values with mean/mode imputation, removing outliers using the NumPy 3-sigma rule, and eliminating duplicate rows.
- Added an auto-generated before/after audit report (DatasetSnapshot) that summarizes every change made to a dataset, cutting manual dataset review time from hours to seconds.
- Wrote the tool to be reusable across arbitrary CSV datasets rather than one-off scripts, so it plugs into any data-cleaning step of a larger pipeline.

EDUCATION

B.Tech in Computer Science & Engineering

2023 – 2027

Indore Institute of Science & Technology (IIST), affiliated to RGPV | CGPA: 7.98 / 10

Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Machine Learning

CERTIFICATIONS

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional — Oracle (Sep 2025)
- AI Agents Intensive Course — Google (Dec 2025)
- Build Your Generative AI Productivity Skills with Microsoft and LinkedIn — Microsoft (Apr 2025)
- SQL and Relational Databases 101 — IBM (Oct 2025)

EXTRACURRICULARS & OPEN SOURCE

- Maintain an active GitHub profile with 29+ public repositories spanning backend systems, GenAI tools, and web apps; solved 150+ DSA problems (Blind 75) across Arrays, HashMaps, Two Pointers, and Linked Lists.